UFCEP6-0-3 · Advanced Information Systems Development

Planning evidence pack

Plans, task records, sprint reviews and risk records supporting the evaluation of the project plan

Item	Detail
Module	UFCEP6-0-3 Advanced Information Systems Development
Assessment	Presentation (50%), second release
Deliverable	Section 3, Planning evidence: initial and revised plans, task records, sprint reviews and risk records
Group	max, finn, henry, peter
Repository	https://github.com/XieQiBin0490/camunda-hospital-referral-group
Submitted version	tag p6-submission
Date	22 October 2026

Table 1. Document control.

1 What this pack is, and what it is not

This pack gathers the planning evidence behind the evaluation on slides 12 and 13 of the presentation. It is deliberately a collection of records rather than a second narrative: where a record exists in the repository it is cited, and where a value is uncertain that is stated rather than smoothed over.

It is not a second portfolio. The portfolio already contains the project and test plan at length; this pack exists because the specification asks for the planning evidence to be identifiable on its own, and because an evaluator should be able to check a claim about the plan without reading 31 pages of task documentation.

Three things in this pack were reconstructed rather than recorded at the time, and each is marked: the risk register (section 6), the SR3 record (section 5, taken from the sprint review notes), and the two sprint windows that differ between the workbook and the submission map (section 3). Everything else is a dated artefact in the repository.

2 Release identity and version control

The specification asks for an identifiable initial release and revised version. Three annotated tags provide that, and every figure in this pack names the tag it belongs to.

Tag	Commit	Points at	What it is used for in this pack
sr1	95bf652	SR1 increment, 17 September 2026	The seven-file increment presented at SR1: folder skeleton, hospital-referral-final.bpmn and the modelling justification
sr2	772d928	Initial release snapshot, 29 September 2026	318 files including 68 Camunda Forms and the 26 September acceptance evidence; the baseline for every “before” value
p6-submission	= `git show p6-submission`	Submitted version	The third revision: 69 forms, 7 process definitions, role-based access enabled, H2 persistence

Table 2. Version identifiers. Each is checkable with git show, and none describes a tag that does not exist.

The release boundary matters for reading this pack: the initial release is sr2 and every figure attributed to September is measured on sr2. The second-release figures are measured on p6-submission. Where a figure changed between the two, both values are given, because a single number would hide the change the second release made.

3 The plans

Four sprints were planned. Two belong to the initial release and two to the second release; this pack records all four, because the evaluation on slide 13 compares the second-release plan with what was delivered, and that comparison only means something against the same capacity assumption used throughout.

Sprint	Planned window	Items selected	Planned hours	Nominal capacity
Sprint 1	14–21 September 2026	17	128 h	80 h
Sprint 2	22–28 September 2026	18	128 h	80 h
Sprint 3	13–19 October 2026	see note	116 h	80 h
Sprint 4	20–22 October 2026	24	80 h	48 h

Table 3. Sprint windows and estimates, read from the backlog workbook. Nominal capacity is 4 people × 4 h per day × the sprint's working days, the assumption used unchanged across all four sprints.

Two notes on the table, both stated rather than resolved:

- Sprint 3 item count. The workbook records 25 selected items summing to 154 hours of estimates; the Sprint 3 task list in the agile delivery record identifies 20 items (S3-01 to S3-20) and a 116-hour plan. The two are not reconciled in the source records. The 116-hour figure is the one used in the evaluation because it is the figure committed at sprint planning; the 154-hour figure is left in the record as an unexplained difference rather than edited out.
- Sprint windows. The workbook gives Sprint 1 as 14–21 September and Sprint 2 as 22–28 September. SUBMISSION_MAP.md gives 14–18 and 21–25. The workbook dates are used here; if the module team published different dates, theirs take precedence and this table should be corrected.

3.1 Sprint goals

Sprint	Goal agreed at planning	Delivered
Sprint 1	Scope, backlog, shared repository, an initial executable BPMN increment and the first forms	One executable model, deployed as a single definition, plus the repository and backlogs
Sprint 2	An integrated increment: the operational model running with the worker fleet and the forms	sr2: four models, 68 forms, 26 worker source files, the 26 September acceptance run
Sprint 3	Draft strategic BPMN and draft i* models with a first requirements trace, reviewable at SR3	v0.9, presented with 3 of the 6 review goals met
Sprint 4	Integrate the fragments, correct the defects found in them, validate and deploy the final models, complete both evaluations and hand in	v1.0: the second release, both evaluations and the submitted repository state

Table 4. Sprint goals and what was actually presented at each review.

4 Task records

Every backlog item carries a first owner, a second owner, an estimate in story points and hours, its dependencies, its acceptance conditions and a status. The second-owner rule is the point of the record: the second owner keeps sufficiently up to date to continue the item if the first cannot deliver, and the two handovers in the project were both exercised under it.

4.1 Initial release, selected items

Item	Task	Owner (1st / 2nd)	Acceptance condition	Status
S1-05	Executable operational model, deployed and linted	max / peter	0 moddle warnings, 0 Camunda 8 lint reports, one definition deploys	Done
S2-08	Rebuild the three fragments as one model family	max / henry	One participant register; every external organisation appears once	Done
S2-12	Generate and bind the Camunda Forms	peter / finn	Every user task that needs a form has one; 99	Done

Item	Task	Owner (1st / 2nd)	Acceptance condition	Status
			of 99 tasks bound	
S2-15	Worker fleet for the automated steps	finn / henry	13 job types; every declared type has a subscriber; contract test passes	Done
S2-19	Acceptance harness and execution	peter / finn	18 cases execute against the deployed model with retained evidence	Done

Table 5. Selected initial-release items. The full S1 and S2 backlogs are in the backlog workbook.

4.2 Second release, the items that carry the increment

Item	Task	Owner (1st / 2nd)	Acceptance condition	Status
S4-01	Triage every defect found in the Sprint 1/2 models	peter / max	Each of the 16 defect categories is fixed or explicitly superseded	Done
S4-05	Rebuild the three fragments as one integrated model family	max / henry	S1, S2, S3 share one participant register; O1 merges all three at task level	Done
S4-08	Deploy the models and re-validate with the Camunda toolchain	peter / max	0 moddle warnings, 0 lint reports, 0 structural defects per model; all definitions deploy	Done
S4-12/13	Complete the i* SD and SR models	finn / henry	SD: 16 actors, 40 typed dependencies. SR: 7 actors, 61 elements, 9 boundary dependencies	Done
S4-16	Alignment evaluation with a full traceability table	henry / max	44 requirements traced to a strategic activity, an i* element and an implementation artefact	Done
S4-18/19	Acceptance criteria review and test execution	peter / finn	All 20 original criteria assessed; 28 revised; 18 cases executed with retained evidence	Done
S4-21	Assemble the portfolio	henry / max	Every required deliverable present, named to convention and linked to its evidence	Done
S4-22/23	Database persistence and role-based access	finn / peter	Records survive a restart; a user with no role sees no tasks	Done
S4-24	Automated RBAC regression suite	peter / finn	Role and audit checks run without manual inspection	**Not done — capacity**

Table 6. Selected second-release items. S4-24 is the only item on this list not delivered; the reason and its consequence are in section 5.4 and in the feedback response record.

Two handovers are recorded in the task records rather than only in conversation. The traceability table moved from max to henry at SU1, which is why it finished in Sprint 2 rather than Sprint 1. The acceptance evidence moved to finn as second owner at SU2 when peter's blocker closed. Both are visible in the owner columns above.

4.3 Standups SU3 and SU4 (individual records)

The two marked standups of this submission are recorded below in the form the specification asks for: what each member reported as complete, the next action they committed to, and any blocker, dependency or handover. Both standups served both units; the substance, not the attendance, is what was marked.

SU3 - 15 October 2026, 09:30

- max. Complete: the level 1 landscape (24 nodes, 7 external participants) and a draft abstraction note. Next: the six-lane referral-to-authorisation model and the strategic-to-operational map. Blocker: the external participant names depend on finn's actor list. Handover: henry is second owner on the landscape file.

- finn. Complete: draft SD model, 12 actors and 27 typed dependencies, rendered from the model source. Next: SR decompositions and the soft goals. Blocker: the renderer still misplaces the resource-dependency arrowhead, so the SVG export is held. Handover: henry is second owner on the SR work.
- henry. Complete: the requirement list closed at 44 identifiers with a source sentence each; traceability at 35 of 44 rows. Next: close the nine open rows, then draft the alignment narrative. Blocker: the open rows are the non-functional requirements, which cannot be closed until the model boundary is agreed. Handover: max is second owner on the traceability table.
- peter. Complete: the revised criteria table with the original wording retained, and a repeatable lint and audit export. Next: run the lint audit over the three strategic files and prepare the SR3 acceptance slide. Blocker: one Camunda 8 error remains - a message boundary event without a subscription correlation key - proposed as a Sprint 4 item rather than patched mid-sprint.

SU4 - 21 October 2026, 09:30

- max. Complete: the three-fragment integration into one model family, the message-flow register and the correlation-key fix. Next: the abstraction note and the final exports. Dependency: the exports are not declared final until peter's lint result is in. Handover: henry is second owner on the abstraction note.
- finn. Complete: SD v1.0 (16 actors, 40 typed dependencies), the soft goals with their conflicts recorded, and the assumptions reconciled. Next: finish SR coverage and co-sign the SD/SR consistency checklist. Dependency: the alignment evaluation needs the final SD element identifiers.
- henry. Complete: traceability complete at 44 identifiers - 28 fully supported, 12 partly, 2 not supported, 2 out of scope - each with a strategic activity and an implementation artefact; the ten prioritised improvements; portfolio assembly started. Next: the alignment narrative, then the hand-in check. Blocker: assembly needs the frozen i* PDFs and the final exports.
- peter. Complete: deployment and re-validation of all four models, and execution of the acceptance cases. Next: the criteria quality check and the change log, then the hand-in check. Blocker resolved: deployment found three XSD sequence violations and the form-binding defect, and all four were fixed and re-verified. Handover: finn is second owner on the test evidence.

What the two standups changed. The traceability table moved from max to henry at SU3 and finished in Sprint 2 rather than Sprint 1. The acceptance evidence moved to finn as second owner at SU4 when peter's blocker closed. The urgency question was raised at both standups and did not move, so it was re-scoped as a dependency on a decision the hospital has not made rather than carried as open work. Both handovers are visible in the first- and second-owner columns of section 4.

5 Sprint reviews

Four reviews, each marked against the increment available on the day. The specification is explicit that later improvements do not retrospectively replace those marks, so nothing in this section has been written back from what the repository looks like now.

5.1 SR1 — 17 September 2026

Presented: sr1, seven files. The model opened, linted cleanly in the Modeler and deployed as a single definition. There were no forms and no worker code, so nothing beyond the model could be demonstrated.

Against the goal: partially met. The executable increment existed and the plan existed as a product backlog plus two sprint backlogs. Configuration management existed as commits and nothing else — no tags, a gap since closed by tagging sr1 and sr2 so both reviews can be reopened as they were.

Three decisions taken on the day:

- Split the pathway across three strategic views and one merged operational model rather than one large diagram.
- Keep the original acceptance criteria next to any revision, so a later improvement cannot quietly hide an earlier failure.
- Treat the pool-naming defect as a rebuild of the model family rather than a patch.

5.2 SR2 — 28 September 2026

Presented: sr2, 318 files. Four models, 68 forms, thirty worker source files and the acceptance evidence. The operational model was driven from the start message to a confirmed treatment schedule with the fleet running, and a second instance was opened on the declined-payment path to show a failure branch rather than describe one.

Against the goal: two of three parts complete. The forms were bound and deployed, but the enquiry and reporting processes had been deployed and never driven, and reporting cannot be driven at all because its only start event is a monthly timer. There was no role-based access control and no persisted store at that point.

Point raised	Agreed response	Backlog change	Owner
Implement database integration to support data persistence	Put an embedded store behind the existing store classes so no worker is rewritten; prove it by writing in one process and reading in another	New item, first in the next increment	finn / henry
Implement role-based access control	Enable authorisation, create the ten candidate groups and one user per group, verify visibility rather than documenting the design	New item; needs a cluster restart to verify	peter / finn
Resolve the issues in the BPMN model so that it accurately represents the business process	Triage every defect in the fragments, fix at source, re-validate all four models	Rebuild item enlarged; each defect gets an ID	max / henry
Implement appropriate form validation to prevent missing or invalid inputs	Required and format validation on every field a gateway reads, plus the fields the case study says must be recorded	New item	peter / henry
Use appropriate input field types for the data being collected	Audit every form against the gateway it feeds and replace free text with selects and numbers wherever the process compares a fixed value	New item; the audit feeds the change log	peter / finn

Table 7. SR2 feedback and the backlog change it produced. Recorded as FB-01 to FB-05 in the portfolio's feedback action record.

5.3 SR3 — 19 October 2026

Presented: v0.9. Draft level 1 and level 2 strategic models, a draft SD model, the first traceability table and the revised acceptance criteria.

Against the goal: three of six goals met. The strategic models and the draft SD model were complete; the SR model, the alignment narrative and the acceptance evidence were not. This is the single largest delivery difference in the project and it is analysed in section 7.

Decision taken at SR3	Consequence
Split the strategic view across three models rather than one, and document the abstraction level explicitly	The abstraction note became a prerequisite for all later modelling work, which cost Sprint 3 its first half
Retain the original acceptance criteria alongside the revisions	The evaluation can show original failures and revisions side by side instead of replacing one with the other
Treat the pool-naming inconsistency as a defect to fix by rebuild rather than by patch	S4-05 was created as a two-owner item; it is the largest single item in Sprint 4

Table 8. SR3 decisions and what each one cost or bought. Taken from the sprint review notes.

5.4 SR4 — 22 October 2026

Status at the time of writing: scheduled. The review falls on the submission date, so this record lists what will be presented and is completed at the session rather than before it. Nothing below asserts an outcome that has not happened.

To be presented	Evidence it will be checked against
The five changes with before-and-after comparison against sr2	Slide 6 to 12; defect log DEF-17 to DEF-21; form field change log
Role-based access, verified rather than designed	rbac/task-visibility.json: 0 tasks without a role, 785 with; 0 failures
Persistence across two JVMs	persistence/run-1-write.txt and run-2-read.txt: 7 of 7 checks
The items not delivered, and why	S4-24 and the two unimplemented feedback points, with reason and residual risk
Planned against actual for Sprints 3 and 4	Section 7 of this pack

Table 9. SR4 agenda. This table is a plan, not a record, and is replaced with the outcome at the session.

6 Risk record

The risk register is the Risks & Assumptions sheet of the backlog workbook. The register below is the version reconstructed for this pack: the identifiers are new, but each row records a risk that was raised in a sprint planning session and the outcome column states what actually happened. Three of the first four materialised.

6.1 Initial release

ID	Risk	Chance	Effect	Response	Owner	Outcome
R-01	The Sprint 1 and 2 fragments were written at different times and will not merge cleanly	High	High	Rebuild rather than patch, with an item and an estimate of its own	max	Materialised. Two files used Zeebe, the third Camunda 7 attributes; identifiers were reused with different meanings. The merge consumed the first half of Sprint 2
R-02	A form is not part of the model and must be deployed separately	Medium	High	Deploy model and forms in one deployment; verify the binding in a test rather than by eye	peter	Materialised. A misspelt form reference was silently ignored and every instance raised FORM_NOT_FOUND — a day lost
R-03	The cluster is not available when the review needs it	Medium	High	Rehearse the run the day before and retain the log; present the recorded run if the cluster refuses to start	peter	Materialised twice. The contingency was used both times, which is why the acceptance evidence is a recorded run
R-04	The urgency rules are never agreed	High	Medium	Model urgency as a soft goal and carry the rule set as a dependency on a decision the hospital has not made	henry	Materialised and still open. No rule set was issued before hand-in

Table 10. Initial-release risks and outcomes. Three of four materialised; R-04 remains open.

6.2 Second release

ID	Risk	Chance	Effect	Response	Owner	Outcome
R-05	Defects that no review finds are only visible at deployment	High	High	Deploy to a real cluster every sprint; treat deployment as a review step, not a release step	peter	Materialised. Three XSD sequence violations and a form-binding defect appeared only when an instance started
R-06	Sprint 4 is planned above the capacity that exists	Certain	High	Prioritise; accept that the lowest-value items will be dropped rather than partially built	henry	Materialised. 80 hours were planned against 48 nominal; S4-24 was dropped rather than half-built
R-07	The embedded database will not permit concurrent writers in the sandboxed environment	Medium	Medium	Verify persistence by write-then-exit-then-read in two JVMs rather than by two concurrent writers; record the	finn	Materialised. The H2 file lock cannot be acquired here; the database opens read-only for a second

ID	Risk	Chance	Effect	Response	Owner	Outcome
				limit		process. Recorded as a stated limit, not hidden

Table 11. Second-release risks and outcomes. The common thread is that the expensive defects were the ones only execution could find.

One risk was deliberately not taken: the i* modelling sat outside the critical path by design, because it depends on nothing the operational model needs. That decision is why it could be completed at the end of the project without moving the release date — and it is also why the i* models cannot be presented as having informed the original design. They were developed afterwards and are used as an evaluation lens.

7 Planned against actual

The table below is the numeric basis for slide 13. Both sprints were planned above the capacity that existed, and both delivered by using the last two days for integration rather than for new work.

Sprint	Planned	Nominal capacity	Over nominal	Delivered	Goals met
Sprint 3	116 h	80 h	+36 h	v0.9	3 of 6
Sprint 4	80 h	48 h	+32 h	v1.0	all of the goal, two items dropped
Both	**196 h**	**128 h**	**+68 h**		

Table 12. Planned hours against nominal capacity for the two second-release sprints. "Over nominal" is planned hours minus capacity, not actual minus planned: actual hours were not recorded consistently enough to make that comparison, and a second estimate dressed as a measurement would be worse than no figure.

7.1 Two causes

The first cause was an abstraction level agreed late. Sprint 3 spent its first half deciding how many strategic models there should be. The consequence was not that less was built; it was that the SR model, the alignment narrative and the acceptance evidence — three of the six review goals — all landed in Sprint 4 instead.

The second cause was a class of defect that no plan could have estimated, because it was invisible until an instance started. Three XSD sequence violations and a form-binding defect appeared only at deployment. They were not in the plan because the plan assumed the Modeler's parse was sufficient evidence that a model was finished. It was not: the Modeler's parser is lenient exactly where the engine's validator is strict.

7.2 What that cost, and what it bought

What it cost: S4-24, the automated RBAC regression suite, was dropped for capacity rather than half-built. Role-based access works and is evidenced by a live two-role check, but it cannot be proved automatically to keep working. The final integration landed in the last two days of Sprint 4, which is the same failure the Sprint 2 retrospective had already identified.

What it bought: the deployment rehearsal produced four defects that no amount of diagram review had found, and closing them is the strongest single piece of evidence in the second release. Reporting them and then fixing them is more defensible than a test plan that only ever ran against stubs.

7.3 Retrospectives

Sprint	Went well	Did not go well	Carried forward
Sprint 1	The abstraction level was agreed in writing before anyone drew anything; the second-owner rule held and the traceability handover from max to henry happened without a gap	128 hours were planned against 80 of capacity, which made the sprint look late from day one while it was going to plan; the merge was left out of the backlog entirely	Agree the model family structure before more BPMN; add the merge as an item with its own estimate and owner
Sprint 2	Deploying to a real cluster was the most useful single step; writing the acceptance harness before the exception paths were specified made each case after the third cheap	Integration landed in the last two days again; the forms were finished last, so the first deployment had nothing to bind to; documentation was treated as the work that happens after the work	A deployment rehearsal at mid-sprint; freeze the model before generating forms; give documentation its own backlog items
Sprint 3	The strategic models were produced quickly once the	The SR model slipped and the i* assumptions note contradicted	Agree the abstraction level in writing before modelling; write the

Sprint	Went well	Did not go well	Carried forward
	abstraction level was agreed; the second-owner rule held	itself once	assumptions note in one sitting; start the acceptance review earlier
Sprint 4	Deployment found four defects that the round-trip, the linter and the Modeler had all passed; reporting them honestly and closing them with a clean run was more defensible than a plan that only ran against stubs	The sprint was planned at 80 hours against 48 nominal and integration again landed in the last two days	Rehearse integration at mid-sprint; build the test harness before the exception paths are specified; treat documentation as scheduled work

Table 13. Retrospectives. The same two findings recur in three of the four sprints, which is the strongest evidence that the plan was wrong in a consistent way rather than unlucky.

8 Where the raw records live

Record	Location	Used in
Product backlog, sprint backlogs, Risks & Assumptions sheet	01_Agile_Planning/ Hospital_Patient_System_Project_Backlog_….xlsx	Sections 3, 4, 6
Task record, contribution matrix, definition of done	01_Agile_Planning/work-division.md	Section 4
Version snapshots and the tag convention	01_Agile_Planning/version-snapshots.md	Section 2
Acceptance run, per-case traces and instance keys	05_Test_Evidence/acceptance-run.json, acceptance-run.log	Section 5, slide 11
Acceptance test plan and criteria	05_Test_Evidence/Acceptance_Test_Plan.md	Section 5
Defect log, including the four found at deployment	08_Repository_Documents/defect-log.md	Sections 4, 7
Form field change log	08_Repository_Documents/form-field-change-log.md	Section 4
Database integration record	08_Repository_Documents/database-integration.md	Section 6
Role-based access configuration and verification	08_Repository_Documents/rbac-setup.md; 05_Test_Evidence/rbac/	Sections 4, 5
Feedback response record	UFCEP6_Feedback_Response_Record.xlsx	Sections 5, 7
Presentation	UFCEP6_Slides.pptx / .pdf	Slides 12, 13

Table 14. Evidence index. Each row is a path a reader can open, not a description of a document that might exist.

9 What this pack does not settle

Four things in the planning record are unresolved, and leaving them visible is deliberate. An evaluator who finds them unstated would be right to assume they were hidden.

Open item	Why it is unresolved	What it changes
Sprint 3 item count: 25 in the workbook against 20 in the task list	The two source records disagree and neither has a date that makes one clearly later	The count is not used in the evaluation; only planned hours are
Sprint 1 and 2 windows differ between the workbook and SUBMISSION_MAP.md	The module team's published dates were not captured in a single place	Neither figure is used in any conclusion; only the sprint order is
Actual hours per sprint	Hours were estimated and committed but not logged per task consistently	The evaluation compares plan with capacity, not plan with actual
S4-24 and the two unimplemented feedback points	Dropped for capacity, not rejected	Recorded with reason and residual risk; carried as the first items of a further release

Table 15. Unresolved items in the planning record.